

1.1 User Scenarios List

The following table provides a concise summary of the four user scenarios covered in this document. Each scenario represents a core workflow performed by hotel staff members using the Hotel Management System.

Scenario ID	User Story	Actor	Goal	Priority
SC-09	UC_09 - View Reservation Dashboard	Staff	Monitor all guest bookings and room occupancy in one central view	High
SC-10	UC_10 - Confirm Pending Reservation	Staff	Approve pending reservations to update guest-visible status	High
SC-11	UC_11 - Register Guest Check - In	Staff	Process check-in and mark room as occupied	High
SC-12	UC_12 - Process Guest Check - Out	Staff	Process check-out, trigger housekeeping, revoke guest access	High

1.2 User Scenarios Extended

US_09: View Reservation Dashboard

1. Receptionist logs in to the system
2. System automatically loads the Reservation Dashboard
3. Dashboard displays all reservations with: Guest Name, Room Number, Check-In Date, Check-Out Date, Current Status
4. Receptionist can filter reservations by date, status, or room number
5. Receptionist can sort the list by any column header
6. Receptionist selects a specific reservation to view full details
7. Receptionist can navigate directly to check-in or check-out process from the dashboard

US_10: Confirm Pending Reservations

1. Receptionist navigates to the Reservation Dashboard
 2. System displays all reservations including those with Pending status
 3. Receptionist filters by status = Pending to identify unconfirmed bookings
 4. Receptionist selects a pending reservation to review full details
 5. System displays guest info, room, dates and payment status
 6. Receptionist clicks the "Confirm" button
 7. System validates room availability for the requested dates
 8. System updates reservation status from Pending to Confirmed
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9. System sends real-time notification to the guest (push and/or email)
10. Dashboard refreshes to reflect the updated status

US_11: Register Guest Check-In

1. Receptionist navigates to the Check-In section of the dashboard
2. Receptionist searches for reservation by guest name, email, or reservation ID
3. System displays confirmed reservations scheduled for today's check-in date
4. Receptionist selects the correct reservation and verifies guest identity
5. Receptionist clicks the "Register Check-In" button
6. System validates: reservation status is Confirmed, room is Available, date matches today
7. System updates reservation status to Checked In
8. System updates room status to Occupied across the entire system
9. System generates and assigns a digital access token to the guest
10. System records the check-in timestamp
11. If room is not yet cleaned: system alerts receptionist and offers to place guest on early check-in waitlist

US_12: Process Guest Check-Out

1. Receptionist opens the active reservations list from the dashboard
 2. Receptionist searches for the guest by name or room number
 3. System displays the guest's reservation and full bill summary: room charges and all ordered services
 4. Receptionist reviews the final bill with the guest
 5. Receptionist processes payment or confirms payment on file
 6. Receptionist clicks the "Process Check-Out" button
 7. System updates reservation status to Checked Out
 8. System revokes the guest's digital access token immediately
 9. System updates room status to Requires Cleaning
 10. System automatically creates and assigns a housekeeping task for the room
 11. System records check-out timestamp and generates receipt
 12. Receipt is delivered to the guest via email or printed copy
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1.3 User Cases

UC_09: View Reservation Dashboard

UC Name	View Reservation Dashboard
Dependency	US_01 (User logs in)
Actors	Receptionist
Preconditions	1. Receptionist is logged in to the system 2. Receptionist has Front Desk or Manager role 3. At least one reservation exists in the system
Description of Main Sequence	1. Receptionist selects "Reservations" from the navigation menu 2. System retrieves all reservation records from the database 3. System displays a table listing: Guest Name, Room Number, Check-In Date, Check-Out Date, Current Status 4. Receptionist optionally applies filters (date range, status, room number) 5. System re-renders the list based on applied filters 6. Receptionist selects a specific reservation to view full details 7. Receptionist can proceed to check-in or check-out directly from the dashboard
Description of Alternative Sequence	1. No reservations match the filters: a. System displays message: "No reservations found for the selected criteria" b. Receptionist adjusts or clears the filters 2. System fails to load data: a. System displays error: "Unable to load reservations. Please try again" b. Receptionist clicks Retry
Non-functional requirements	1. Dashboard must load within 2 seconds under normal conditions 2. Dashboard reflects real-time status changes without manual page refresh 3. Access is restricted to authenticated staff with appropriate role 4. System must support at least 100 concurrent staff sessions
Postconditions	1. Dashboard is displayed with all relevant booking data 2. No reservation records are modified by viewing 3. Receptionist has full visibility to proceed with any reservation action

UC_10: Confirm Pending Reservations

UC Name	Confirm Pending Reservations
Dependency	US_01 (User logs in)UC_09 (View Reservation Dashboard)
Actors	Receptionist, Guest (notified)
Preconditions	1. Receptionist is logged in with reservation management permissions 2. At least one reservation exists with status = Pending 3. The reservation's check-in date has not yet passed
Description of Main Sequence	1. Receptionist opens the Reservation Dashboard and filters by Pending status 2. Receptionist selects a pending reservation to view full details 3. System displays: guest info, room, dates, payment status 4. Receptionist clicks the "Confirm" button 5. System validates that the room is still available for the requested dates 6. System updates reservation status from Pending to Confirmed 7. System sends real-time notification to the guest (push and/or email) 8. Guest dashboard reflects updated status within 5 seconds 9. Receptionist dashboard refreshes to show Confirmed status
Description of Alternative Sequence	1. Room no longer available (race condition): a. System alerts: "Room no longer available for these dates" b. Receptionist reassigns room or rejects the reservation 2. Receptionist rejects the reservation: a. Receptionist clicks "Reject" and enters a mandatory reason b. System updates status to Rejected c. System notifies the guest of the rejection with reason provided 3. Guest account is deactivated: a. System flags the issue and prevents confirmation b. Receptionist escalates to manager to resolve
Non-functional requirements	1. Status update must complete within 3 seconds 2. Guest notification must be delivered within 5 seconds of confirmation

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	3. All status changes must be logged with receptionist ID and timestamp 4. System must prevent double-booking confirmation (race condition handling)
Postconditions	1. Reservation status = Confirmed 2. Guest receives confirmation notification 3. Guest dashboard reflects updated status in real-time 4. Room is held and no longer bookable for overlapping dates

UC_11: Register Guest Check-In

UC Name	Register Guest Check-In
Dependency	US_01 (User logs in)UC_09 (View Reservation Dashboard)UC_10 (Confirm Pending Reservations)
Actors	Receptionist, Guest
Preconditions	1. Receptionist is logged in with check-in permissions 2. Reservation status = Confirmed 3. Check-in date = today (or within allowed early check-in window) 4. Room status = Available
Description of Main Sequence	1. Receptionist navigates to the Check-In section of the dashboard 2. Receptionist searches for reservation by guest name, email or reservation ID 3. System displays confirmed reservations scheduled for today 4. Receptionist selects the correct reservation and verifies guest identity 5. Receptionist clicks "Register Check-In" 6. System validates: reservation is Confirmed, room is Available, date is valid 7. System updates reservation status to Checked In 8. System updates room status to Occupied 9. System generates and assigns digital access token to the guest 10. System records check-in timestamp in the reservation record
Description of Alternative Sequence	1. Room is not yet cleaned (status = Requires Cleaning): a. System alerts: "Room not ready for check-in" b. Receptionist places guest on early check-in waitlist

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	c. Guest is notified automatically when room becomes available (US-08) 2. Guest arrives on incorrect date: a. System shows date mismatch warning b. Receptionist requests manager override c. Manager approves and override is logged with reason 3. Room status = Maintenance: a. System prevents check-in b. Receptionist reassigns guest to another available room 4. Guest already checked in: a. System displays: "Guest is already checked in" b. Receptionist reviews record to resolve the duplicate
Non-functional requirements	1. Check-in process must complete within 3 seconds 2. Digital access token must be generated at the moment of check-in 3. Room status must update in real-time across all system views 4. Check-in is only permitted for reservations with status Confirmed
Postconditions	1. Reservation status = Checked In 2. Room status = Occupied 3. Digital access token generated and assigned to guest 4. Check-in timestamp recorded in the system

UC_12: Process Guest Check-Out

UC Name	Process Guest Check-Out
Dependency	US_01 (User logs in)UC_09 (View Reservation Dashboard)UC_11 (Register Guest Check-In)
Actors	Receptionist, Guest, Housekeeper (receives task)
Preconditions	1. Receptionist is logged in with check-out permissions 2. Reservation status = Checked In 3. All room charges and service orders are recorded in the system
Description of Main Sequence	1. Receptionist opens active reservations list and selects the guest's reservation 2. System displays full bill: room charges, ordered services, extras 3. Receptionist reviews the bill with the guest

Hotel Management System Requirements Specification

	4. Receptionist processes payment or confirms payment on file 5. Receptionist clicks "Process Check-Out" 6. System updates reservation status to Checked Out 7. System revokes guest's digital access token immediately 8. System updates room status to Requires Cleaning 9. System automatically creates and assigns housekeeping task for the room 10. System records check-out timestamp 11. System generates receipt and delivers to guest by email or print
Description of Alternative Sequence	1. Guest requests late check-out: a. Receptionist verifies room is not pre-booked for later that day b. If available: receptionist extends check-out time in the system c. System updates the check-out timestamp accordingly 2. Guest disputes a charge: a. Receptionist marks the disputed charge in the system b. Dispute is escalated to manager c. Check-out proceeds; bill is flagged as partially disputed 3. Payment fails: a. System prevents check-out completion b. Receptionist resolves payment before proceeding 4. Housekeeping system is offline: a. Housekeeping task is queued locally b. Task is delivered to housekeeper once connection is restored c. Room is still marked Requires Cleaning in the system
Non-functional requirements	1. Digital access token must be revoked within 60 seconds of check-out confirmation 2. Housekeeping task must be auto-triggered — manual creation is not required 3. Room must not accept a new check-in until housekeeper marks it Available 4. All check-out actions must be logged with receptionist ID and timestamp 5. If housekeeping system is offline, task must be queued and delivered on reconnection
Postconditions	1. Reservation status = Checked Out 2. Guest digital access token revoked 3. Room status = Requires Cleaning 4. Housekeeping task created and sent to assigned housekeeper 5. Receipt generated and delivered to guest

Hotel Management System Requirements Specification

6. Check-out timestamp recorded in the system

1.4 Functional Requirements

1. UC_09 – View Reservation Dashboard

Req #	Requirement	Comments	SME Reviewed
FR_09	The system must provide a Reservation Dashboard accessible to authenticated staff immediately after login, displaying all reservations with guest name, room number, check-in date, check-out date and current status.	Critical for daily front-desk operations.	Front Desk Staff
FR_10	Staff must be able to filter the reservation list by date range, status and room number, and sort by any column header. The list must re-render in real-time based on applied filters without a page reload.	Enables quick lookup during busy periods.	Front Desk Staff
FR_11	Staff must be able to select any reservation from the dashboard to view its full details and navigate directly to the check-in or check-out workflow from that record.	Supports seamless front-desk workflow transitions.	Front Desk Staff

2. UC_10 – Confirm Pending Reservations

Req #	Requirement	Comments	SME Reviewed
FR_14	Staff must be able to confirm a reservation with status = Pending. Upon confirmation, the system must validate	Core receptionist workflow. Must prevent double-booking via race condition handling.	Front Desk Staff

Hotel Management System Requirements Specification

	room availability for the requested dates and update the reservation status from Pending to Confirmed.		
FR_15	Upon confirmation, the system must send a real-time notification to the guest (push and/or email) and update the guest dashboard to reflect the Confirmed status within 5 seconds.	Guest must be informed immediately.	Guest Services
FR_16	Staff must be able to reject a pending reservation by clicking Reject and entering a mandatory reason. The system must update the status to Rejected and notify the guest with the reason provided.	Prevents silent rejections; audit trail required.	Front Desk Staff

3. UC_11 – Register Guest Check-In

Req #	Requirement	Comments	SME Reviewed
FR_21	Staff must be able to register a guest check-in by searching for a confirmed reservation (by guest name, email or reservation ID) and clicking Register Check-In. The system must validate that the reservation is Confirmed, the room is Available and the date is valid before proceeding.	Core front desk operation. Only reservations with today's check-in date must be shown in the check-in view.	Front Desk Staff
FR_22	Upon successful check-in, the system must update the reservation status to Checked In, update the room status to Occupied across the entire system in real-time, and record the check-in timestamp.	Room occupancy must be reflected instantly system-wide.	System Admin
FR_23	The system must generate and assign a	Required for keyless room access. Waitlist	System Admin

Hotel Management System Requirements Specification

	cryptographically secure digital access token to the guest at the moment of check-in. If the room is not in Available status, the system must alert staff and offer to place the guest on an early check-in waitlist.	supports early-arrival guest experience.	
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4. UC_12 – Process Guest Check-Out

Req #	Requirement	Comments	SME Reviewed
FR_28	Staff must be able to process guest check-out by selecting an active Checked In reservation, reviewing the full bill (room charges and all service orders) with the guest, processing payment or confirming payment on file, and clicking Process Check-Out.	Core front desk operation. Payment failure must block check-out completion.	Front Desk Staff
FR_29	Upon check-out, the system must immediately revoke the guest's digital access token, update the room status to Requires Cleaning, and automatically create and assign a housekeeping task to the responsible housekeeper. The room must not accept a new check-in until the housekeeper marks it as Available.	Eliminates manual task creation. Security and housekeeping workflow requirement.	Housekeeping / System Admin
FR_30	The system must generate a payment receipt upon check-out and deliver it to the guest by email or printed copy. The system must record the check-out timestamp. If the housekeeping system is offline, the housekeeping task must be queued locally and	Required for financial record keeping and system resilience.	Billing / System Admin

Hotel Management System Requirements Specification

	delivered once the connection is restored.		
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1.4b Non-Functional Requirements

1. UC_09 – View Reservation Dashboard

Req #	Requirement	Comments	SME Reviewed
NFR_09a	The Reservation Dashboard must fully load within 2 seconds under normal operating conditions and must support at least 100 concurrent authenticated staff sessions without performance degradation.	Performance baseline for busy periods.	System Admin
NFR_09b	The dashboard must reflect reservation status changes in real-time without requiring a manual page refresh, using live data synchronisation.	Prevents staff acting on stale data.	System Admin
NFR_09c	Access to the Reservation Dashboard is restricted to authenticated staff with Front Desk or Manager role. Unauthorised access attempts must display: "You do not have permission to perform this action".	Role-based access control requirement.	System Admin

2. UC_10 – Confirm Pending Reservations

Req #	Requirement	Comments	SME Reviewed
NFR_10a	Reservation status updates (Pending to Confirmed or Rejected) must complete within 3 seconds of the staff action and be reflected	User experience and data consistency requirement.	System Admin

Hotel Management System Requirements Specification

	on both the staff and guest dashboards.		
NFR_10b	The system must prevent double-booking: if a race condition is detected during confirmation, the system must alert staff that the room is no longer available and block the confirmation.	Data integrity requirement.	System Admin
NFR_10c	All reservation status changes must be logged with the staff member ID, timestamp, and the old and new status values for audit purposes.	Required for audit trail and compliance.	System Admin

3. UC_11 – Register Guest Check-In

Req #	Requirement	Comments	SME Reviewed
NFR_11a	The check-in process must complete end-to-end within 3 seconds of the staff clicking Register Check-In, including status updates and token generation.	Operational speed requirement for busy front desks.	System Admin
NFR_11b	The digital access token must be generated and assigned to the guest at the exact moment of check-in confirmation. Token generation must not require a separate manual step.	Security and automation requirement.	System Admin
NFR_11c	Room status must update to Occupied in real-time across all system views (dashboard, room management, housekeeping) within 3 seconds of check-in confirmation. Check-in is only permitted for reservations with status = Confirmed.	Prevents assigning an unready or already-occupied room.	System Admin

4. UC_12 – Process Guest Check-Out

Hotel Management System Requirements Specification

Req #	Requirement	Comments	SME Reviewed
NFR_12a	The guest's digital access token must be revoked within 60 seconds of check-out confirmation. All check-out actions must be logged with the staff member ID and timestamp.	Security requirement. Immediate revocation prevents unauthorised room access post-checkout.	System Admin
NFR_12b	The housekeeping task must be auto-triggered upon check-out without any manual creation step. If the housekeeping system is offline, the task must be queued and delivered automatically on reconnection, and the room must still be marked Requires Cleaning.	System resilience and automation requirement.	Housekeeping / System Admin
NFR_12c	The check-out workflow must be completable in no more than 5 user interactions from the dashboard. Status changes must propagate across all system views within 3 seconds of the check-out action.	Efficiency and UX consistency requirement.	System Admin

1.5 Use Case Diagram

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Use Case Diagrams: UC_09 · UC_10 · UC_11 · UC_12

